

Applied analytics & data systems — portfolio summary

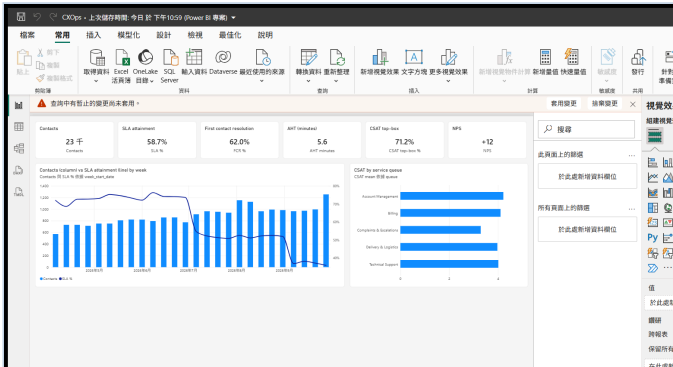
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MSc in Data Science (Artificial Intelligence Application), University of Macau · BSc Civil Engineering · Macau SAR selection for the 46th WorldSkills Competition (Business Software Solutions), 2022



1 · CUSTOMER-SERVICE OPERATIONS DASHBOARD — POWER BI, SQL, PYTHON

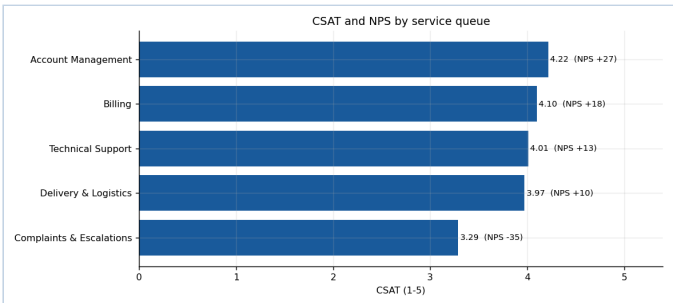
- Designed a **star-schema model** over 23,162 contacts and 15,721 cases (26 weeks, five channels), with a date table and six dimensions.
- Authored **36 DAX measures** covering SLA attainment, first-contact resolution, handle time, backlog ageing, CSAT/CES/NPS and cost per contact — each definition documented with its pitfalls.
- Added **30 automated data-quality checks** (key integrity, Likert ranges, flag coherence, calendar coverage) that gate a refresh instead of silently publishing wrong numbers.
- Every figure is validated against an **independent pandas implementation**, so the dashboard numbers are provably reproducible.



Finding	Evidence
Phone wait time doubled from week 14	CES 5.09 → 3.98 (−1.11, $p < .001$); CSAT 4.18 → 3.37
Email first-response SLA collapsed from week 23	80.6% → 23.9% ($p \approx 1e-296$); open backlog concentrated in 7–30 day bucket
Volume spike was <i>not</i> an experience problem	Billing contacts 153 → 302/week with CSAT flat (4.01 vs 4.11)

2 · CUSTOMER-EXPERIENCE METRICS & STATISTICAL ANALYSIS — SPSS, PYTHON

- Built an **SPSS-native dataset** (.sav with variable labels, value labels and measurement levels) plus full **SPSS syntax** for NPS/CES/CSAT, bootstrap confidence intervals, chi-square, Welch t-tests, ANOVA with Tukey post-hoc, Kruskal-Wallis, standardised-beta driver analysis and ordinal logistic regression (PLUM).
- NPS +11.6** (41.7% promoters, 30.1% detractors), bootstrap 95% CI [+9.1, +14.0] — resampled because NPS is a difference of proportions.
- First-contact resolution is the dominant driver of CSAT** (standardised $\beta = +0.49$, $R^2 = 0.347$), ahead of escalation (−0.24), transfers (−0.11) and wait time (−0.04).
- Channel and NPS group are **not independent** ($\chi^2 = 209.7$, $df\ 8$, $p < .001$, Cramér's $V = 0.157$).



- **Methodological control:** pooling wait time across channels suggests longer waits raise FCR (+0.39, $p < .001$); tested within channel the effect disappears and reverses for live chat (−0.12, $p = .002$) — a channel confound that would have produced the wrong recommendation.

Deliverables: 4-page bilingual findings report, SPSS .sav + .sps, and a procedure-by-procedure SPSS $\rightleftharpoons$ Python equivalence table. No SPSS licence was available on this machine, so the syntax is shipped with expected values and cross-checked numerically in Python — stated openly rather than faked.

HOW THIS MAPS TO THE ROLE

Requirement	Evidence in this portfolio
Develop and maintain databases and data systems	Star schema (7 tables), SQL data-quality gates, reproducible pipeline
Automate workflows for real-time analytics and performance tracking	Automated model/report generation from source files; 30 checks that block a bad refresh
Statistical tools for diagnosis and prediction	Bootstrap CIs, chi-square, Welch t-tests, ANOVA + Tukey, Kruskal-Wallis, regression, ordinal logistic
NPS, CES and satisfaction surveys	Full metric definitions, response-rate and non-response handling, driver analysis
Power BI, SPSS, Python, SQL, Excel/VBA	Used throughout; measure definitions, SPSS syntax and .sav files all included

Disclosure. All data is synthetic, generated by a seeded, documented generator; the projects demonstrate method and tooling, not any organisation's real performance. Every reported figure is reproducible by running the included scripts. Code, documentation and outputs are available on request or in a repository.